FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

# CRIME IQ

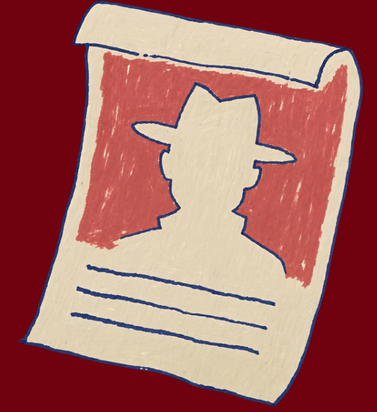
CRIME IQ

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# ABSTRACT





CRIMEIQ IS AN INTERACTIVE AI-BASED CRIME INVESTIGATION GAME DESIGNED TO DEVELOP PROBLEM-SOLVING AND DECISION-MAKING SKILLS.




IT PRESENTS FICTIONAL CRIME CASES WITH SUSPECTS, EVIDENCE, WITNESS STATEMENTS, AND CRIME SCENE DETAILS.



PLAYERS USE CLASSICAL AI ALGORITHMS TO EXPLORE CLUES, EVALUATE SUSPECTS, AND IDENTIFY THE CULPRIT.

THUS, THE GAME PROVIDES A HANDS-ON LEARNING EXPERIENCE, DEMONSTRATING HOW AI TECHNIQUES SUPPORT LOGICAL REASONING AND EFFECTIVE DECISION-MAKING.

# PROBLEM STATEMENT



## WHY CRIME IQ?



**6.24M**

*CRIMES REGISTERED IN 2023*



**17,100/DAY**

*AVERAGE CASES REGISTERED*



**15.85L**

*CASES PENDING FROM PREVIOUS YEARS*



**40%+**

*OF CERTAIN INVESTIGATIONS PENDING >6 MONTHS*

**TOO MANY CASES + TOO MUCH INFORMATION + TOO MANY POSSIBLE LEADS → SLOWER INVESTIGATION AND DIFFICULT DECISION MAKING.**

**HENCE, THERE IS A NEED TO TURN COMPLEX CASE INFORMATION INTO CLEAR, PRIORITIZED INVESTIGATION ACTIONS.**

# LITERATURE SURVEY

## OATLEY ET AL. (2007)

**Data Mining**

Applied data mining to extract useful patterns from crime and forensic data.

## KEPPENS & ZELEZNIKOW (2011)

**Rule-Based + Bayesian Reasoning**

Developed an integrated reasoning framework combining evidence to evaluate plausible crime scenarios.

## WANG ET AL. (2023)

**Bayesian Network**

Used Bayesian reasoning to guide investigation directions and assess evidence importance.

## FARBER (2025)

**AI Image Analysis**

Applied AI image analysis to accelerate crime-scene evidence screening and interpretation.

## HAIGH & DLUGOLENSKI (2026)

**AI Suspect Identification**

Explored AI-based identification methods to assist police in identifying potential suspects more efficiently.

# ALGORITHMS

01



## Iterative Deepening Search (IDS)

Explores investigation leads progressively by searching at increasing depths, helping identify relevant connections between people, locations, and evidence.

02



## A* Search

Finds the most promising investigation path by considering both the cost of an action and its estimated usefulness.

03



## Constraint Satisfaction Problem (CSP) + Backtracking

Applies multiple evidence-based constraints to filter potential suspects and revisits earlier choices when a constraint is violated.

04



## Propositional Logic

Uses IF–THEN rules to represent relationships between evidence and derive logical conclusions during an investigation.

05



## Forward Chaining

Starts with known case facts and repeatedly applies rules to derive new conclusions and determine investigation priorities.

# RESEARCH GAP

## 01

### FRAGMENTED SOLUTIONS

Existing AI research often focuses on specific investigation tasks such as evidence analysis, suspect investigation, or forensic analysis.

## 02

### LIMITED ACTION PRIORITIZATION

Less emphasis on systematically exploring alternative leads and selecting the next investigation action.

## 03

### NEED FOR INTEGRATION

Scope for a unified workflow connecting lead exploration, suspect filtering, evidence reasoning, and action selection.

### OUR RESEARCH GAP

**From analyzing case information → to supporting investigation decision making.**

# RESEARCH OBJECTIVE

CRIMEIQ

AI CRIMINAL INVESTIGATION
DECISION SUPPORT SYSTEM

## 01

### ANALYZE & STRUCTURE CASE INFORMATION

Take diverse case information and represent it in a structured form that the AI can reason over.

- Ingest case data (evidence, suspects, witnesses, locations, etc.)
- Identify and model relationships between entities
- Structure information for efficient reasoning

## 02

### EXPLORE & EVALUATE INVESTIGATION POSSIBILITIES

Explore connected leads, apply constraints, and evaluate alternative investigation paths to identify the most promising leads.

- Explore connected leads
- Apply constraints to eliminate unsuitable candidates
- Evaluate alternative investigation paths
- Identify leads/actions that deserve attention

## 03

### SUPPORT INVESTIGATION DECISION MAKING

Convert the analysis into prioritized suspects/leads and recommended next actions, with evidence-based reasoning.

- Prioritize suspects/leads
- Recommend next investigation actions
- Provide evidence-based reasoning for recommendations
- Assist investigators in decision making (not an autonomous decision-maker)

OUR GOAL

**FROM COMPLEX CASE INFORMATION TO SMARTER INVESTIGATION DECISIONS.**

# ARCHITECTURE

## 01 USER INTERFACE



HTML5 • Vanilla CSS3 • Vanilla JavaScript (ES6+)
Case screens • Suspects • Clues • Timeline • Accusation

## 02 CASE DATA & GAME LOGIC



JavaScript case files
Suspects • Evidence • Locations • Alibis • Timeline

## 03 CLASSICAL AI ALGORITHM ENGINE



JavaScript

- **Iterative Deepening Search (IDS)** → Clue exploration
- **A* Search** → Lead prioritization
- **CSP + Backtracking** → Test suspect hypotheses
- **Propositional Logic** → Evaluate evidence rules
- **Forward Chaining** → Derive conclusions

## 04 GAME INTERACTION & SCORING



Suspect/evidence panels • Graph investigation board • Algorithm traces
Accusation and score screens

## 05 LEARNING OUTCOME



Apply classical AI algorithms to solve interactive crime mysteries.

## TECH STACK

**FRONTEND + NATIVE JAVASCRIPT**

HTML5 • CSS3 • Vanilla JavaScript • Node.js • HTML Canvas • SVG • Web Audio API • Git/GitHub

# CONCLUSION




CRIMEIQ
AI CRIMINAL INVESTIGATION GAME

## 01



### OBJECTIVE ACHIEVED

CrimeIQ successfully demonstrates how classical AI algorithms can be applied through an interactive game to solve crime investigations.

## 02



### LEARNING OUTCOMES

The project provides a hands-on learning experience, helping users understand AI concepts through realistic and engaging scenarios.

## 03





### OVERALL IMPACT

CrimeIQ serves as an educational tool that makes AI learning more interactive, interesting and practical through a real-world inspired theme.

OUR CONCLUSION

**CRIMEIQ SUCCESSFULLY ACHIEVES ITS GOAL OF PROVIDING AN INTERACTIVE AND ENGAGING PLATFORM TO LEARN AND APPLY AI TECHNIQUES FOR CRIMINAL INVESTIGATION SCENARIOS.**

# FUTURE ENHANCEMENTS



01

## MORE CASES

Add new crime mysteries with varied difficulty levels.

02

## SMARTER AI FEATURES

Improve clue analysis and combine algorithm outputs for richer reasoning.

03

## BETTER GAMEPLAY

Add hints, progress tracking, and player performance scores.

OUR GOAL

**A richer, more challenging AI learning experience.**

# THANK YOU